Quiz 3 (*Oct 25, Marks: 10, Duration: 30 mins*) **COL703/7203, Jul-Nov 2026**

Name: **Entry No.:**

1. **[5 marks]** Use natural deduction to prove the following.

$$\forall x \exists y\ (P(x) \vee Q(y)) \quad \vdash \quad \exists y \forall x\ (P(x) \vee Q(y))$$

2. Consider a predicate logic formula $\phi := \psi_1 \wedge \psi_2 \wedge \psi_3$, where

ψ_1: $\forall x \exists y\ R(x, y)$
ψ_2: $\forall x\ \neg R(x, x)$
ψ_3: $\forall x \forall y \forall z\ (R(x, y) \wedge R(y, z) \rightarrow R(x, z))$

   (a) **[2 marks]** Is ϕ satisfiable? Justify your answer.

   (b) **[3 marks]** Can ϕ have a finite model (i.e., an assignment where the universe has only finitely many elements)? Give such a finite model, or argue otherwise.